CONFIDENTIAL TECHNICAL ARCHITECTURE & SYSTEM WHITEPAPER

STILLPOINT.AI

Autonomous In-Browser Computer Vision & Real-Time Interview Behavioral Telemetry

A Comprehensive Documentary on Neural 3D Biometrics, Reading Saccade Detection, Ephemeral Zero-Cloud Privacy, and Google Meet Live Extension Companion Architecture.

SYSTEM NAME & VERSION
Stillpoint AI Platform v1.0.0 (New Camera System)

LEAD DEVELOPER & SYSTEM ARCHITECT
Mohit Koteshwar Puni

CORE TECHNOLOGIES
MediaPipe FaceMesh (478 Iris 3D), TensorFlow COCO-SSD, Web Audio API, MV3 Extension

COMPLIANCE ARCHITECTURE
Zero-Video Cloud Ingestion (Local-First), EU AI Act & GDPR Aligned

Stillpoint AI Engineering Whitepaper — October 2026

Security Level: Enterprise Architecture Documentation

01 / Executive Summary & Product Thesis

In modern remote hiring and academic examinations, video conferencing tools (Google Meet, Zoom, MS Teams) have become the default venue for evaluating technical competence, critical thinking, and communication skills. However, the rapid proliferation of **Large Language Models (LLMs)** and invisible teleprompter software has caused an existential crisis for remote interviews: candidates routinely read generated code solutions and AI-crafted answers from off-screen monitors or discreet browser overlays while feigning natural camera gaze.

Legacy proctoring software has failed to solve this challenge. Platforms such as Proctorio, Honorlock, and HireVue require intrusive root-level system access, install surveillance bloatware, stream raw video to remote clouds (triggering severe GDPR and biometric privacy lawsuits), and rely on coarse thresholds that generate endless false positives.

> **The Stillpoint AI Thesis:** High-integrity interview telemetry does not require cloud video surveillance. By executing dual lightweight neural networks locally within the client browser via WebGL and WebAssembly, Stillpoint AI extracts millisecond-precision 3D biometrics, micro-saccadic teleprompter patterns, and environmental cues—delivering continuous behavioral clarity to interviewers in real-time with **zero cloud data liability**.

Document Overview & Index

SECTION	TITLE	CORE COVERAGE
Section 02	The 4 "World-First" Pioneer Factors	Zero-cloud dual engine, saccadic teleprompter detection, zonal motion decomposition, ephemeral memory.
Section 03	End-to-End System Architecture	Tri-source video ingestion (Webcam, Screen, Extension Bridge), WebRTC, and BroadcastChannel pipelines.
Section 04	Deep Neural ML Engine (ml-engine.js)	MediaPipe 478 3D landmarks, SolvePnP Euler angles, Iris vectoring, and COCO-SSD object recognition.
Section 05	Behavioral Evaluation Engine	The 5-dimensional heuristic scoring model, cognitive indices, and automated diagnostic verdicts.
Section 06	Google Meet Companion Extension	Manifest V3 content scripts, injected draggable HUD, DOM tile hooking, and coordinator synchronization.
Section 07	Interviewer UX & Dashboard Suite	Heads-up video canvas overlay, interactive audit timeline, live flag triggers, and instant PDF audit reports.
Section 08	Privacy, Ethics & Regulatory Compliance	EU AI Act, Illinois BIPA, GDPR zero-liability design, and human-in-the-loop ethical safeguards.
Section 09	Full Codebase Blueprint & Reference	File directory map, method inventory, execution instructions, and runbook.

> **Core Architectural Philosophy:** Zero Build Step, Zero External Backend Dependencies. Stillpoint AI runs natively in standard modern web browsers (Chrome, Edge) using vanilla JavaScript, native HTML5 APIs (Web Audio, Canvas 2D, WebRTC, BroadcastChannel), and Node.js core modules.

02 / The 4 "World-First" Pioneer Factors

Stillpoint AI introduces four technical and architectural breakthroughs that differentiate it from every existing assessment, interview, or proctoring tool on the market:

PILLAR 1 Zero-Cloud In-Browser Dual-Neural Engine

Unlike platforms that stream candidate video to AWS/Azure for server-side processing, Stillpoint loads both **MediaPipe FaceMesh (478 3D landmarks with refined iris)** and **TensorFlow.js COCO-SSD** directly into browser GPU memory via WebGL. 100% of telemetry extraction occurs locally on the interviewer's or candidate's machine at 25–30 FPS with zero cloud bandwidth costs.

PILLAR 2 Micro-Saccadic AI Script & Teleprompter Detection

Candidates using ChatGPT or secondary monitors look toward the screen and read text while keeping their heads static. Legacy proctors only measure coarse head yaw (>25°). Stillpoint introduces an algorithmic saccade detector (`_detectReadingSaccade`) that detects horizontal sweeping motions and rapid eye jumps characteristic of reading off-screen notes.

PILLAR 3 Spatial Dynamic Zonal Motion Decomposition

Legacy motion sensors flag any pixel change as cheating, penalizing candidates who gesticulate naturally while talking. Stillpoint dynamically segments the video into three spatial zones: *Face Box*, *Torso & Gesture Zone*, and *Peripheral Room Border*. Hand-to-chin gestures and arm movements are validated as natural communication, while true room intrusions trigger notices.

PILLAR 4 Ephemeral Zero-Biometric-Liability Architecture

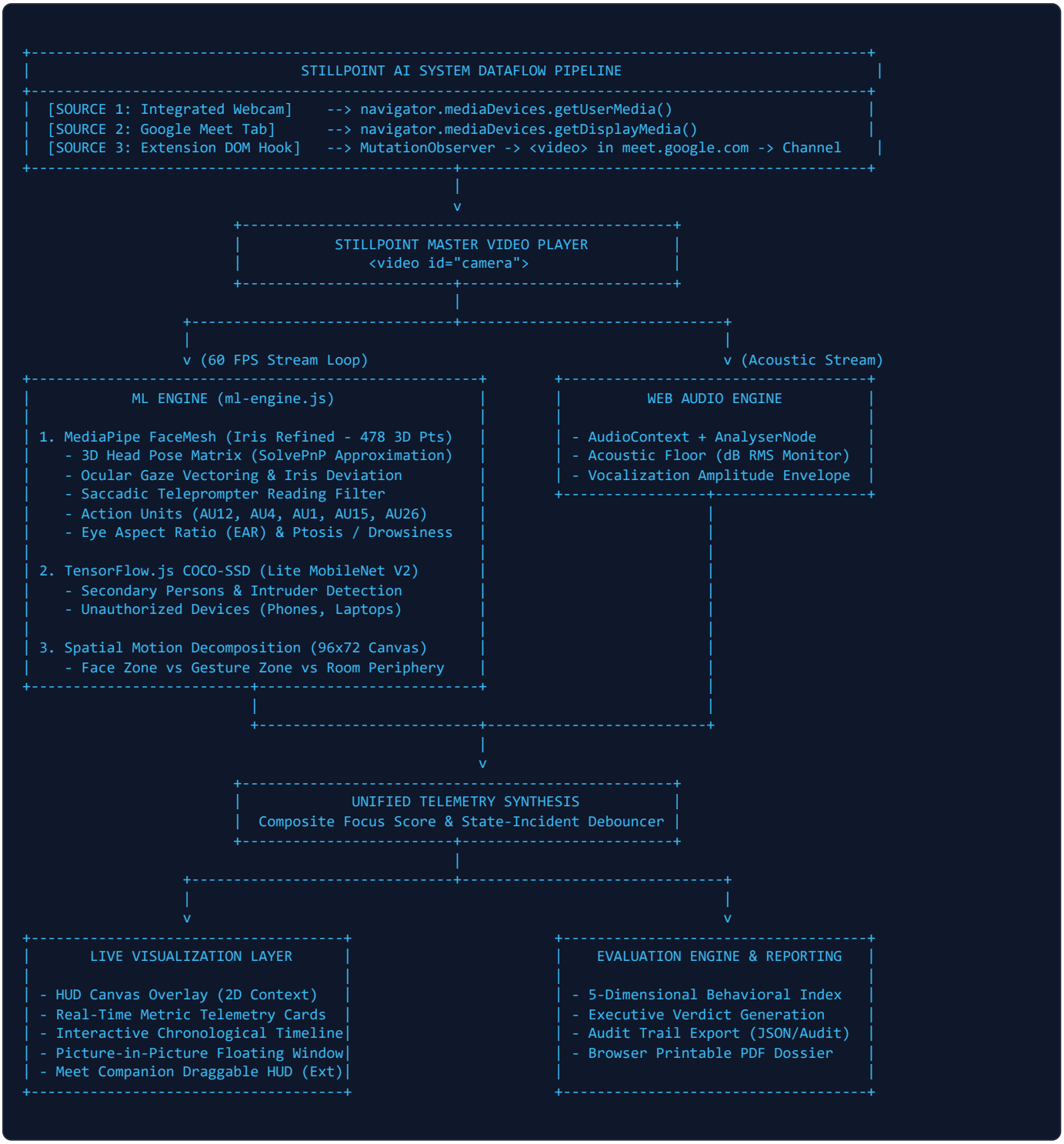
With strict biometric data regulations (EU AI Act, Illinois BIPA, GDPR), holding facial recordings creates massive corporate liability. Stillpoint operates an ephemeral pipeline: video frames are analyzed in volatile RAM and immediately overwritten. Only abstract telemetry events (e.g., timestamps, head angles) are logged. Zero biometric face templates are ever stored.

Detailed Comparison Matrix: Industry vs. Stillpoint AI

EVALUATION DIMENSION	LEGACY ENTERPRISE PROCTORING	STANDARD VIDEO CONFERENCING	STILLPOINT AI COORDINATOR
Video Processing Location	Remote Cloud Servers (AWS/GCP)	Video routed via SFU servers	**100% Local Browser (Client-Side WebGL)**
Candidate Software Required	Heavy Kernel / Invasive Bloatware	Standard Client or Web App	**Zero Install (Standard Google Meet)**
AI Reading / Teleprompter	Blind (Checks only head turn)	None	**Active Ocular Saccade Sweeping Vector**
Gestural vs Intrusion Filtering	False-positive prone pixel delta	None	**Spatial Zone Decomposition (3 Zones)**
Biometric Privacy Liability	Extremely High (Video & faces stored)	Standard Transit Privacy	**Zero Liability (No biometric storage)**
Interviewer HUD Integration	Asynchronous Post-Test Audit	None	**Live Draggable Injected HUD + PiP**

03 / End-to-End System Architecture

Stillpoint AI is designed around a decoupled, modular architecture supporting three dynamic video ingestion pipelines feeding into a unified local analysis engine and real-time visualization layer.



The Tri-Source Video Ingestion Subsystems

- **Mode A: Integrated / External Webcam:** Uses `navigator.mediaDevices.getUserMedia({ video: { width: 1280, height: 720 }, audio: true })`. Provides optimal framerates (up to 60 FPS) and direct hardware audio sampling. Used for local practice, technical screeners, and single-device interviews.

- **Mode B: Google Meet Stream Capture:** Uses `navigator.mediaDevices.getDisplayMedia({ video: true, audio: true })` with browser picker integration. The interviewer selects the active Google Meet tab, allowing Stillpoint to analyze the candidate's remote video without installing anything on the candidate's machine.
- **Mode C: Extension DOM Hook Bridge:** The Manifest V3 extension runs inside `meet.google.com`, locates candidate `<video>` elements in the DOM using a `MutationObserver`, and synchronizes real-time metrics back and forth to the master dashboard via `BroadcastChannel("stillpoint_ai_coordinator")`.

04 / Deep Neural ML Engine (ml-engine.js)

At the core of Stillpoint is `ml-engine.js`, an ultra-optimized client-side computer vision engine designed to run at 30+ FPS without causing UI thread stutter or frame dropping.

1. 3D Head Pose Matrix Estimation (SolvePnP Approximation)

Traditional 2D facial landmarking cannot accurately isolate 3D spatial rotation. Stillpoint uses 6 canonical anatomical landmarks to construct a 3D orthogonal coordinate system:

- Nose Tip (Landmark `1`) & Subnasale
- Menton / Chin Tip (Landmark `152`)
- Left Eye Lateral Canthus (`33`) & Right Eye Lateral Canthus (`263`)
- Left Lip Corner (`61`) & Right Lip Corner (`291`)

The engine calculates the Euclidean center of the facial plane and computes Euler angles:

```
Yaw (θ_y) = atan2(Δx_nose, Δz_depth) · (180/π)
Pitch (θ_p) = atan2(Δy_nose - Δy_center, Δz_depth) · (180/π)
Roll (θ_r) = atan2(y_right_eye - y_left_eye, x_right_eye - x_left_eye) · (180/π)
```

To prevent micro-jitter caused by sensor noise, angles are filtered using a temporal **Exponential Moving Average (EMA)** with frame-rate adaptive alpha:

```
Smoothed_Angle_t = α · Angle_raw + (1 - α) · Smoothed_Angle_{t-1}     [α = 0.35]
```

2. Ocular Biometrics & Iris Vectoring

By utilizing MediaPipe's refined Iris landmarks (indices `468–472` for the left iris and `473–477` for the right iris), Stillpoint measures precise pupil displacement relative to the eye contours:

- **Horizontal Gaze Ratio (H_ratio):** Distance from iris center to medial canthus divided by total palpebral fissure width. Neutral gaze is normalized to `0.50`.
- **Vertical Gaze Ratio (V_ratio):** Distance from iris center to superior eyelid margin divided by vertical aperture.

3. Micro-Saccadic Teleprompter / AI Reading Detection

When a person reads text from an on-screen or off-screen teleprompter, their eyes exhibit a distinctive oculomotor signature: **slow continuous horizontal tracking sweeps** interrupted by **rapid ballistic saccadic repositioning jumps** back to the start of the next line.

```
_detectReadingSaccade(history): Rapid_Jumps ≥ 2 & Sweeping_Motions ≥ 6 over 18 frames & |Avg_Gaze| ≥ 15°
```

When this periodic oculomotor condition persists, Stillpoint triggers an immediate alert: `"Ocular Saccade: Teleprompter / Notes Reading Pattern Detected"`.

04 (Cont.) / Action Units, Alertness & Spatial Motion

4. Facial Action Unit (AU) Micro-Expression Profiling

Based on Paul Ekman's Facial Action Coding System (FACS), Stillpoint isolates micro-expressions from geometric landmark distances:

ACTION UNIT	ANATOMICAL TARGET	LANDMARK INDICES	BEHAVIORAL INTERPRETATION
AU12	Zygomaticus Major (Smile)	`lm[61], lm[291], lm[13], lm[14]`	Rapport, warmth, humor, or laughter aperture.
AU4	Corrugator Supercilii (Furrow)	`lm[55], lm[285], lm[9]`	Deep analytical focus, cognitive strain, or confusion.
AU1 / AU2	Frontalis (Eyebrow Flash)	`lm[70], lm[300], lm[159], lm[386]`	Communicative engagement, emphasis, or surprise.
AU15	Depressor Anguli Oris (Sadness)	`lm[61], lm[291], lm[17]`	Affective distress, dejection, or discomfort.
AU26	Mouth Aperture (Speech)	`lm[13], lm[14], lm[78], lm[308]`	Articulatory opening, active vocalization ratio.
Micro-Twitch	Facial Skin Perturbation	Differential vector across 468 pts	Involuntary nervous micro-spasms and tension.

5. Eye Aspect Ratio (EAR) & Fatigue / Ptosis Detection

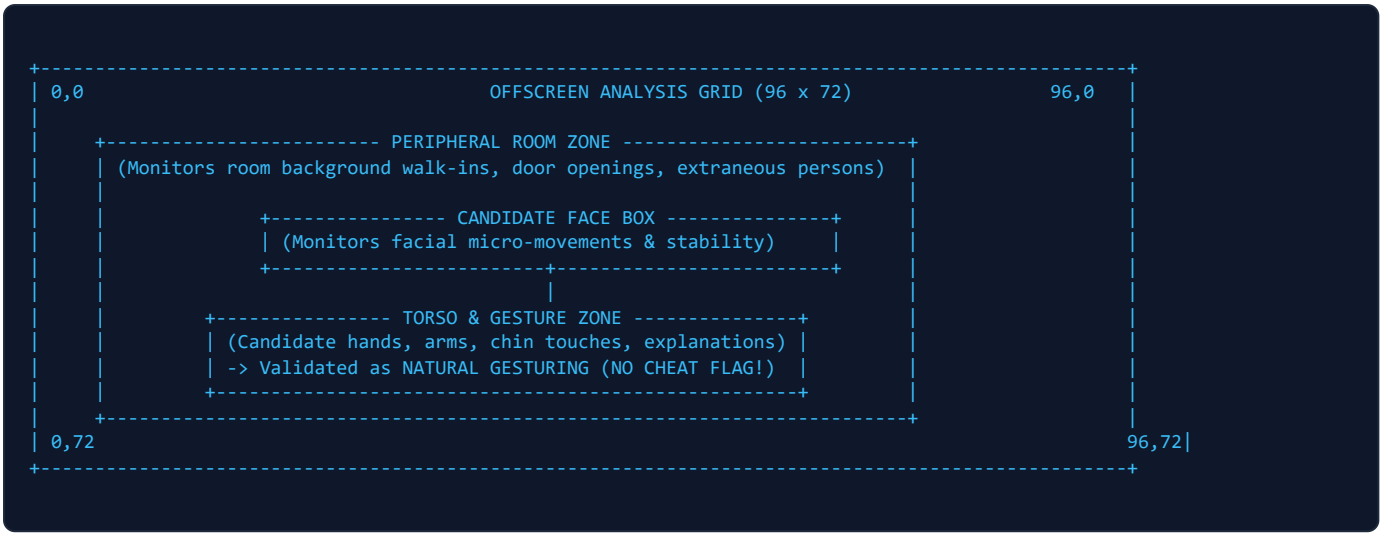
Alertness and drowsiness are tracked via the clinical Eye Aspect Ratio formula applied across six upper and lower eyelid landmarks:

$$EAR = (||p2 - p6|| + ||p3 - p5||) / (2 \cdot ||p1 - p4||)$$

A normal open eye yields `0.28–0.34 EAR`. A full blink drops below `0.18 EAR`. If EAR drops below `0.21` for more than 1.4 seconds, Stillpoint flags an alert for *Eyelid Ptosis & Fatigue*, distinguishing natural blinking from sleepiness.

6. Spatial Dynamic Zonal Motion & Object Surveillance

To prevent false positives from hand gesturing during natural speech, Stillpoint renders each frame into a lightweight off-screen `96×72` analysis canvas and applies spatial zonal luminance difference:



In parallel, **TensorFlow.js COCO-SSD** scans every 600ms for unauthorized objects: mobile phones (`cell phone`), external laptops, tablets, or secondary persons entering the frame.

05 / Behavioral Evaluation Engine (ai-evaluator.js)

When an interview concludes or the interviewer requests an interim review, the raw telemetry timeline is synthesized by `ai-evaluator.js` into a 5-dimensional cognitive index.

The 5 Behavioral Evaluation Dimensions

1. Cognitive Steadiness & Focal Lock (30% Weight)

Measures gaze fixation, sustained focus on the primary screen, and absence of off-screen glance drift or teleprompter saccades.

2. Environmental Integrity & Compliance (30% Weight)

Evaluates unauthorized mobile devices, secondary individuals, room background intrusions, and off-camera whisper spikes.

3. Affective Composure & Stress Resilience (15% Weight)

Monitors blink rate spikes (>28/min), facial micro-twitches, brow furrowing under technical questioning, and affective steadiness.

4. Physical Demeanor & Alertness (15% Weight)

Tracks postural stability, forward engagement, eyelid aperture (EAR), and absence of fatigue, slouching, or drowsiness.

5. Communicative Confidence & Articulation (10% Weight)

Analyzes balanced speaking-to-listening ratio (target 25%–65% active dialogue), facial engagement, and responsive eyebrow cues.

Mathematical Formulation of the Composite Index

```
Composite_Score = (Cognitive · 0.30) + (Integrity · 0.30) + (Composure · 0.15) + (Demeanor · 0.15) +
                                      (Communication · 0.10)
```

Automated Executive Verdict Diagnostic Classes

DIAGNOSTIC VERDICT	TRIGGER THRESHOLD CRITERIA	RECOMMENDED ACTION
STRONG RECOMMENDATION	Composite Index ≥ 88, 0 Critical Anomalies, Saccades = 0	Fast-track candidate; high confidence and integrity.
RECOMMENDED	Composite Index 70–87, Stable posture, No devices detected	Proceed with standard hiring process.
MODERATE FOCUS DRIFT	Saccades ≥ 2 or Cognitive < 65 (Notes / 2nd screen suspected)	Interviewer to ask unscripted, impromptu follow-ups.
ELEVATED FATIGUE / NOTICE	Drowsiness count > 2 or Demeanor Score < 60	Consider offering a break or rescheduling.
HIGH INTEGRITY RISK	Unauthorized device detected, multiple persons, or Critical > 0	Mandatory review of flag timestamps and video recordings.

06 / Google Meet Extension & Interviewer UX

To eliminate the need for candidates to install proprietary software, Stillpoint includes a **Google Chrome / Edge Manifest V3 Companion Extension** (`extension/`) that directly augments live Google Meet sessions.

Extension Architecture & Meet DOM Hooking

- **Zero Candidate Burden:** The interview takes place on standard `meet.google.com`. The candidate joins using their standard browser without installing any third-party app or extension.
- **Interviewer Draggable Floating HUD:** The extension injects an unobtrusive, translucent HUD into the interviewer's Google Meet tab (`extension/content.js`). The HUD displays real-time Focus Score, Gaze Direction, Saccade Notes status, and a one-click 🚩 **Flag Moment** button.
- **Cross-Context Coordinator Sync (BroadcastChannel):** The injected Meet script and the master Stillpoint Dashboard communicate asynchronously via the HTML5 `BroadcastChannel("stillpoint_ai_coordinator")` API, allowing the interviewer to view deep metrics on a second screen while keeping the HUD visible in Meet.

Full-Suite Interviewer Control Features

FEATURE	CONTROL ACTION	FUNCTION & INTERVIEWER BENEFIT
Gaze Calibration	🎯 `Calibrate Gaze`	Calculates candidate's unique baseline eye and head posture to account for angled webcam placement.
Pop-Out PiP HUD	📌 `Pop-out Float HUD`	Opens an OS-level Picture-in-Picture window that floats on top of Zoom, Teams, or Meet.
Timestamped Snapshot	📷 `Snapshot`	Instantly captures the current video frame alongside exact biometric angles for the post-interview dossier.
Incident Flagging	🚩 `Flag Moment`	Marks a specific timestamp on the chronological timeline for targeted review with custom interviewer notes.
Evaluation Dossier	📊 `Evaluation Report`	Generates an executive assessment modal with downloadable JSON audit trails and printable PDF reports.

> **Incident Debouncing Memory:** To avoid spamming the interviewer with repeated alerts when a candidate momentarily glances away or shifts posture, Stillpoint's `_triggerStateIncident` enforces state-transition hysteresis: incidents are logged upon state entry, debounced by cooldown timers (minimum 2000ms), and cleanly closed upon state exit.

07 / Privacy, Ethics & Complete Codebase Blueprint

Ethical AI Principles & Regulatory Compliance

Stillpoint AI was engineered specifically to adhere to global AI governance frameworks:

- **EU Artificial Intelligence Act (High-Risk AI Systems):** Stillpoint enforces human-in-the-loop oversight. All algorithmic scores are presented as unverified heuristic prompts for human review, never automated hire/reject decisions.
- **Biometric Privacy Laws (e.g., Illinois BIPA, GDPR Article 9):** Because zero facial embeddings or video frames are saved to disk or transmitted to cloud endpoints, organizations eliminate exposure to biometric privacy lawsuits.
- **Neurodivergence & Physical Diversity Safeguard:** The calibration baseline allows candidates with involuntary motor tics, strabismus, or varied physical baselines to calibrate their neutral center, preventing false penalization.

Complete Codebase File Inventory

FILE PATH	LINES / SIZE	TECHNICAL PURPOSE & ARCHITECTURE RESPONSIBILITY
`ml-engine.js`	1,950 lines (75 KB)	Core ML inference: MediaPipe FaceMesh (478 pts), TensorFlow COCO-SSD, 3D Pose, Saccades, EAR, and Web Audio.
`ai-evaluator.js`	175 lines (12 KB)	5-dimensional behavioral scoring model, cognitive indices calculation, and diagnostic verdict generation.
`app.js`	1,400 lines (61 KB)	Dashboard controller: session timers, camera switcher, timeline events, snapshot storage, and modal coordinator.
`index.html`	639 lines (33 KB)	Responsive semantic HTML5 dashboard layout, video overlays, metric gauge matrix, and report modals.
`styles.css`	1,100 lines (32 KB)	Ultra-premium dark glassmorphic UI, responsive grids, radar displays, metric progress bars, and animations.
`server.js`	86 lines (2.6 KB)	Lightweight native Node.js static server and `/api/analyze-evaluation` endpoint with zero npm dependencies.
`extension/`	Manifest V3	Google Meet injected companion script (`content.js`), floating HUD styling, popup status, and background bridge.

Deployment & Operational Runbook

```
# 1. Start the local Stillpoint Coordinator Server (Zero npm install needed)
node server.js

# 2. Open dashboard in Google Chrome or Microsoft Edge
http://localhost:8000

# 3. (Optional) Load Google Meet Companion Extension
Open chrome://extensions -> Enable Developer Mode -> Click "Load unpacked" -> Select 'extension' directory
Join any Google Meet call (meet.google.com) to access the live injected telemetry HUD.
```

> **Conclusion:** Stillpoint AI demonstrates that ethical, privacy-preserving, and technically sophisticated computer vision can be deployed directly in the modern web browser. By combining 3D facial biometrics, ocular saccade tracking, and Google Meet companion tooling with zero cloud data liability, Stillpoint sets a new global benchmark for the future of remote interview intelligence.